

Ayesha Fazal Lashkarwala

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SUMMARY

I have hands-on experience building and deploying NLP inference pipelines in a research environment. At TU Berlin, built a multi-agent pipeline for table reasoning with rollback and verification, evaluated vLLM, NVIDIA Triton, and TGI for production throughput, and engineered a FastAPI ML microservice. Currently finishing an MSc in AI at BTU Cottbus. Seeking a role where I can bring this applied ML perspective to a team shipping at scale.

EXPERIENCE

Technische Universität Berlin

Jul. 2025 – Present

Working Student - Data Integration and Data Preparation

Berlin, Germany

- Architected a visual logging dashboard that cut time spent on manual LLM output validation and sped up debugging across experiments
- Evaluated 5+ inference servers (e.g., vLLM, NVIDIA Triton, TGI) to optimize for throughput and latency, authoring a comparative report that streamlined the final deployment decision
- Engineered an ML microservice using FastAPI that served model inference for the team's experimental pipeline
- Integrated Polars for data ingestion, cutting load time on large uploads and standardising raw data into Hugging Face Datasets
- Designed a multi-agent backtracking pipeline for agentic table reasoning, preventing compounding errors from premature row filtering and column selection
- Built an auditor and Critic Agent framework to validate tabular transformations, rolling back execution to the last valid state

Brandenburgische Technische Universität Cottbus - Senftenberg

Oct. 2025 – Sep. 2026

Student Assistant - Explainable Machine Learning (XAI)

Cottbus, Germany

- Developed and delivered a Python programming curriculum to 100+ enrolled students
- Taught core ML concepts, including linear and logistic regression and data hygiene
- Demonstrated impact of Feature Engineering (Standardization, Normalization) on model performance
- Prepared final exam tasks assessing core ML concepts

Spur Solutions Pvt Ltd

Jul. 2023 – Sep. 2024

Associate Software Engineer

Karachi, Pakistan

- Performed in-depth data analysis to standardize and harmonize system metrics, enhancing data consistency and reliability across the software platform
- Reduced execution time by 89% by optimizing a complex stored procedure within the legacy system which supported real-time decision-making based on user-defined rules
- Extended a legacy .NET system to support role-specific workflows, eliminating several manual workarounds previously used by the team
- Integrated SignalR into a legacy .NET system, replacing polling-based updates with real-time communication and reducing perceived latency for end users

Lahore University of Management Sciences (LUMS)

Jan. 2023 – May 2023

Teaching Assistant - Principles and Techniques of Data Science

Lahore, Pakistan

- Ran weekly Python tutorials and office hours for 80+ students
- Designed assignments, graded quizzes, and led a semester-long project from scoping to final evaluation for 80+ students

- Queried user behavior data using SQL and Mixpanel to identify patterns that informed product feature decisions
- Identified and escalated data quality issues that were affecting analytics reliability across the product team

SKILLS

Programming : Python, SQL, C# , C++, JavaScript, R
AI and ML : Machine Learning, PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, XAI (LIME), Generative AI, Agentic AI, Multi-Agent Systems
Backend and Data : FastAPI, Polars, Pandas, ETL Pipelines, REST APIs
Cloud : Docker, vLLM, NVIDIA Triton, TGI **Tools** : Git/GitHub, Mixpanel
Soft Skills : Problem Solving, Pattern Recognition, Communication, Teaching, Leadership, Collaboration, Time Management, Analysis, Research
Language Skills : *English* C2 - Fluent, *German* A2 - Conversational, *Urdu* C2 - Native

PROJECTS

AdaptiRead | *GitHub* Nov. 2025 – Jan. 2026

- Built a containerized, conditional NLP pipeline to simplify text for dyslexic readers
- Engineered a complexity metric using SUBTLEX-UK Zipfian word frequencies and weighted syllable counts
- Fine-tuned a T5-small Seq2Seq Transformer on the WikiAuto dataset for text simplification
- Prevented meaning drift with a DistilBERT guardrail, regenerating text if BERTScore F1 less than 0.85
- Integrated LIME for Explainable AI (XAI) and deployed the PyTorch pipeline via Docker

Driver Facial Expression Recognition using Mobile ViT | *GitHub* Oct. 2025 – Feb. 2026

- Evaluated GC-ViT against MobileViT to optimize real-time driver monitoring
- Achieved an 80% reduction in parameters (from 28M to 5.6M)
- Reduced total inference time by ~3 hours across the dataset
- Validated MobileViT-S on the KMU-FED dataset, maintaining a strong 0.9042 F1-score

Bachelor Thesis - Adaptive Learning Application for Maths | *GitHub* Feb. 2022 – May 2023

- Built a machine learning system to classify common mathematical mistakes made by 220 young learners in low-resourced Pakistani schools, identifying eight mistake categories
- Designed and implemented a machine learning pipeline using classical (Naive Bayes, Logistic Regression) and deep learning techniques, achieving up to 87% accuracy with classical models
- Used Google Vision API for image-to-text conversion and feature extraction, creating detailed CSV files for model training and testing
- Performed data augmentation and image processing (cropping, grayscale adjustments) to expand the dataset, improving deep learning model accuracy from 37% to 60%

Authorship Attribution Model | *GitHub* Nov. 2022 – Dec. 2022

- Scraped tweets of 600+ Twitter/X users
- Performed data cleaning using regular expressions (regex), extracted features forming the vocabulary
- Used Laplace Smoothing to handle out-of-vocabulary words
- Trained and compared KNN, Gradient Boosting, and MLP classifiers; Gradient Boosting performed strongest

EDUCATION

Brandenburgische Technische Universität Cottbus - Senftenberg Oct. 2024 – Nov. 2026
Master of Science in Artificial Intelligence Cottbus, Germany

Lahore University of Management Sciences (LUMS) Sep. 2019 – Jun. 2023
Bachelor of Science in Computer Science Lahore, Pakistan